

A Data-Driven Car Recommendation Model

Objective: What are the most reliable vehicles for each price range, and desired body style?

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Data Sources

Car Body Styles (API): <https://vpic.nhtsa.dot.gov/api/>

Car Prices (CSV): <https://www.kaggle.com/datasets/abdulmalik1518/cars-datasets-2025>

Car Reliability (HTML): <https://repairpal.com/reliability>

Description

Given a price range, intended use case, and reliability, the objective of this project is to create a recommendation tool to help those in search of a new car to make the best possible purchase.

First, the Kaggle dataset will be used to establish a price for each make and model. Next, the vPIC API will be called to pull in body style for each make and model.

Then, reliability scores for each make and model will be extracted by scraping repairpal.com with the following page format: repairpal.com/reliability/make/model/. These scores will be matched to their respective car in the dataset.

After merging all datasets, a model will be implemented that ranks cars by reliability and price given the body type and price range. The final model will generate tailored recommendations for users based on their inputs for price range and vehicle body style.